→ Height of a fuzzy set
$h(A)$

→ Normal fuzzy set
$A(x)$ as normal if $h(A)=1$

→ Sub normal fuzzy set
$A(x)$ is a subnormal if $h(A)<1$

Some standard operations in fuzzy sets.

→ Proper subset
$A(x)$ $B(x)$
$x \in X$
$A(x) \subset B(x)$, if $\mu_A(x) < \mu_B(x)$
$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$
$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$

→ Equal fuzzy set
$A(x) = B(x)$
$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$
$B(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$

→ Compliment of a fuzzy set

$A(x)$ $\quad \bar{A}(x)$

$\bar{A}(x) = 1 - A(x)$

If $A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$

$\bar{A}(x) = \{(x_1, 0.9), (x_2, 0.8), (x_3, 0.7)\}$

→ Intersection of fuzzy set:

$A(x)$ $\quad B(x)$ $\quad x \in X$

$(A \cap B)(x)$

$\mu_{(A \cap B)}(x) = \min\{\mu_A(x), \mu_B(x)\}$

$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$

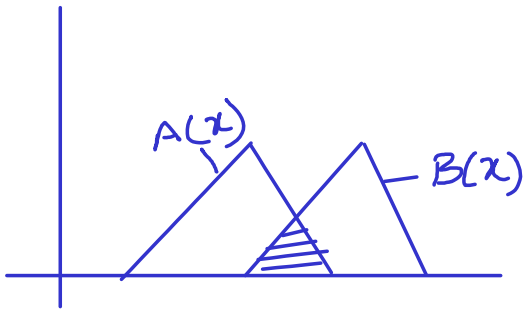
$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$

$\mu_{(A \cap B)} x_1 = \min\{\mu_A(x_1), \mu_B(x_1)\} = 0.1$

$\mu_{(A \cap B)} x_2 = 0.2$

$\mu_{(A \cap B)} x_3 = 0.3$

$$(A \cap B)(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$$



→ Union of fuzzy sets

$$\mu_{(A \cup B)}(x) = \max\{\mu_A(x), \mu_B(x)\}$$

→ Algebraic product of fuzzy sets.

Algebraic product of two fuzzy sets $A(x)$ and $B(x)$ is denoted by $A(x) \cdot B(x)$

$$A(x) \cdot B(x) = \{x, \mu_A(x) \cdot \mu_B(x)), x \in X\}$$

Ex: $A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$

$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$

$$A(x) \cdot B(x) = \{(x_1, 0.05), (x_2, 0.14), (x_3, 0.24)\}$$

→ Multiplication of fuzzy set with a crisp value:

$$d \cdot A(x) = \{ (x, d \times \mu_A(x)), x \in A \}$$

Ex: $A(x) = \{ (x_1, 0.1), (x_2, 0.2), (x_3, 0.3) \}$

If $d = 2$

$$d \cdot A(x) = \{ (x_1, 0.2), (x_2, 0.4), (x_3, 0.6) \}$$

→ Algebraic sum of two fuzzy sets:

$A(x)$ $B(x)$ $x \in X$

$$A(x) + B(x) = \{ (x, \mu_{A+B}(x)), x \in X \}$$

$$\mu_{A+B}(x) = \mu_A(x) + \mu_B(x) - \mu_A(x) \cdot \mu_B(x)$$

$$A(x) = \{ (x_1, 0.1), (x_2, 0.2), (x_3, 0.3) \}$$

$$B(x) = \{ (x_1, 0.5), (x_2, 0.7), (x_3, 0.8) \}$$

$$A(x) + B(x) = \{ (x_1, 0.55), (x_2, 0.76), (x_3, 0.86) \}$$

→ Bounded sum of two fuzzy set $A(x)$ and $B(x)$

Bounded sum is denoted as follows:

$$A(x) \oplus B(x) = \{(x, \mu_{A \oplus B}(x)), x \in X\}$$

Where $\mu_{A \oplus B}(x) = \min\{1, \mu_A(x) + \mu_B(x)\}$

Example:

$$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$$

$$A(x) \oplus B(x) = \{(x_1, 0.6), (x_2, 0.9), (x_3, 1)\}$$

→ Algebraic difference of two fuzzy sets $A(x)$ and $B(x)$ is represented as follows:

$$A(x) - B(x) = \{(x, \mu_{A-B}(x)), x \in X\}$$

$$\mu_{A-B}(x) = \mu_{A \cap \bar{B}}(x)$$

$$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$$

$$\bar{B}(x) = \{(x_1, 0.5), (x_2, 0.3), (x_3, 0.2)\}$$

$$A(x) - B(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.2)\}$$

→ Bounded difference of two fuzzy sets $A(x)$ and $B(x)$ is represented as follows:

$$A(x) \ominus B(x) = \{(x, \mu_{A \ominus B}(x), x \in X\}$$

$$\mu_{A \ominus B} = \max\{0, \mu_A(x) + \mu_B(x) - 1\}$$

Example:

$$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8)\}$$

$$A(x) \ominus B(x) = \{(x_1, 0), (x_2, 0), (x_3, 0.1)\}$$

→ Cartesian product of two fuzzy sets $A(x)$ and $B(y)$ is represented as $A(x) \times B(y)$

$A(x)$ $\quad x \in X$

$B(y)$ $\quad y \in X$

$$\mu_{A \times B}(x, y) = \min(\mu_A(x), \mu_B(y))$$

$$A(x) = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5), (x_4, 0.6)\}$$

$$B(y) = \{(y_1, 0.8), (y_2, 0.6), (y_3, 0.3)\}$$

$$\min(x_1, y_1) = \min(0.2, 0.8) = 0.2$$

$$A \times B = \begin{bmatrix} 0.2 & 0.2 & 0.2 \\ 0.3 & 0.3 & 0.3 \\ 0.5 & 0.5 & 0.3 \\ 0.6 & 0.6 & 0.3 \end{bmatrix}$$

Composition of fuzzy set

$$A = [a_{ij}]$$

$$B = [b_{jk}]$$

$$C = A \circ B$$

$$[c_{ik}] = [a_{ij}] \circ [b_{jk}]$$

$$c_{ik} = \max\left[\min(a_{ij}, b_{jk})\right]$$

$$A = [a_{ij}] = \begin{bmatrix} 0.2 & 0.3 \\ 0.5 & 0.7 \end{bmatrix} \quad B = [b_{jk}] = \begin{bmatrix} 0.3 & 0.6 & 0.7 \\ 0.1 & 0.8 & 0.6 \end{bmatrix}$$

$$c_{11} = \max\left[\min(a_{11}, b_{11}), \min(a_{12}, b_{21})\right]$$

$$= \max\left[\min(0.2, 0.3), \min(0.3, 0.1)\right]$$

$$= 0.2$$

$$A = [a_{ij}] = \begin{bmatrix} 0.2 & 0.3 \\ 0.5 & 0.7 \end{bmatrix} \quad B = [b_{jk}] = \begin{bmatrix} 0.3 & 0.6 & 0.7 \\ 0.1 & 0.8 & 0.6 \end{bmatrix}$$

$$c_{12} = \max\left[\min(a_{11}, b_{12}), \min(a_{12}, b_{22})\right]$$
$$= \max\left[\min(0.2, 0.6), \min(0.3, 0.8)\right]$$
$$= 0.3$$

$$c_{13} = \max\left[\min(0.2, 0.7), \min(0.3, 0.6)\right]$$
$$= 0.3$$

$$c_{21} = \max\left[\min(0.5, 0.3), \min(0.7, 0.1)\right]$$
$$= 0.3$$

$$c_{22} = \max\left[\min(a_{21}, b_{12}), \min(a_{22}, b_{22})\right]$$
$$= \max\left[\min(0.5, 0.6), \min(0.7, 0.8)\right]$$
$$= 0.7$$

$$c_{23} = \max\left[\min(0.5, 0.7), \min(0.7, 0.6)\right]$$
$$= 0.6$$

$$C = \begin{bmatrix} 0.2 & 0.3 & 0.3 \\ 0.3 & 0.7 & 0.6 \end{bmatrix}$$

Properties of fuzzy sets.

1. Law of excluded middle:

In crisp set, $A \cup \bar{A} = X$

However, according to the fuzzy set

$$A \cup \bar{A} \neq X$$

2. Law of contridiction.

As per crisp set, $A \cap \bar{A} = \phi$

However in fuzzy set $A \cap \bar{A} \neq \phi$

Measurement of fuzziness and inaccuracy of fuzzy sets :

$$X = \{x_1, x_2, x_3 \ldots x_n\}$$

$A(x)$

H(A) is the entrophy of the fuzzy set A(x)

$$H(A) = \frac{-1}{n} \sum_{i=1}^{n} \left[\mu_A(x_i) \log\{\mu_A(x_i)\} + \{1-\mu_A(x_i)\} \log\{1-\mu_A(x_i)\} \right]$$

Ex:

$$A(x) = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.4), (x_4, 0.5)\}$$

$$H(A) = -\frac{1}{4} \Big[\{0.1 \times \log(0.1) + 0.9 \times \log(0.9)\} + \{0.3 \log(0.3) + 0.7 \times \log(0.7)\}$$
$$\{0.4 \times \log(0.4) + 0.6 \times \log(0.6)\} + \{0.5 \times \log(0.5) + 0.5 \times \log(0.5)\} \Big]$$

$$= 0.2499$$

Inaccuracy:

A(x) and B(x) are fuzzy sets.

$$I(A;B) = \frac{-1}{n}\sum_{i=1}^{n}\left[\mu_A(x_i)\log(\mu_B(x_i)) + (1-\mu_A(x_i)\log(1-\mu_B(x_i))\right]$$

Ex:

$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3), (x_4, 0.4)\}$

$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8), (x_4, 0.9)\}$

Determine inaccuracy of the fuzzy set B(x) with respect to fuzzy set A(x).

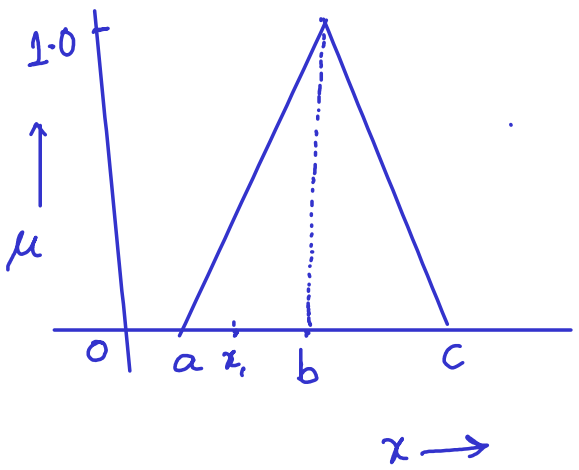
$$I(A;B) = \frac{-1}{4}\Big[\{0.1\times\log(0.5) + 0.9\times\log(0.5)\} + \{0.2\times\log(0.7) + (0.8)\times\log(0.3)\} + \{0.3\times\log(0.8) + 0.7\times\log(0.2)\} + \{0.4\times\log(0.9) + 0.6\times\log(0.1)\}\Big]$$

$$= \underline{0.4717}$$

Triangular membership:

It is denoted Triangle $(x; a, b, c)$

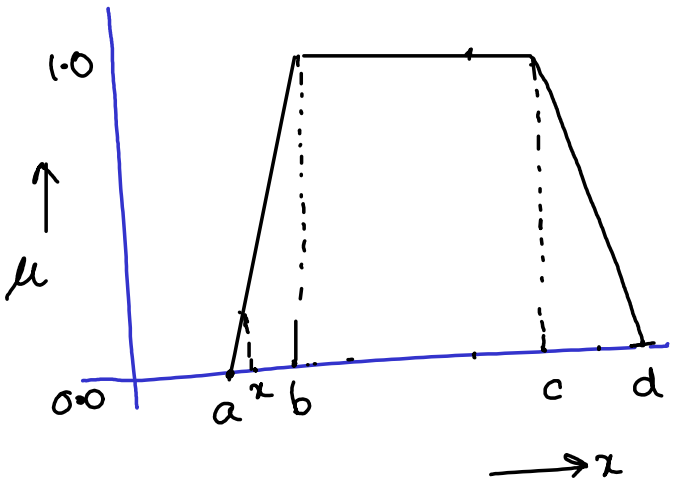
$\mu_{triangle}$



$$\mu_{triangle} = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$

Trapezoidal Membership

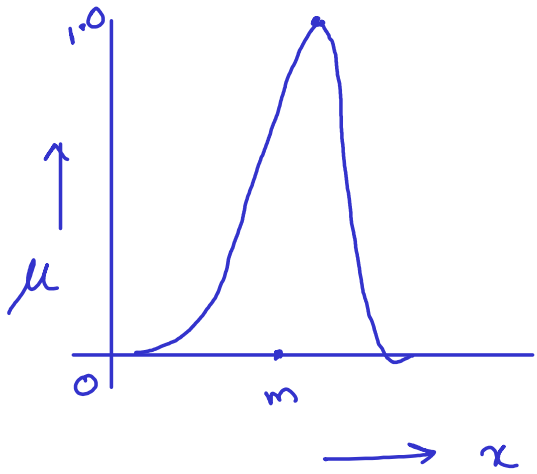
Trapezoidal$(x; a, b, c, d)$



$$\mu_{trapezoidal} = \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right)$$

Gaussian Membership

Gaussian $(x; m, \sigma)$



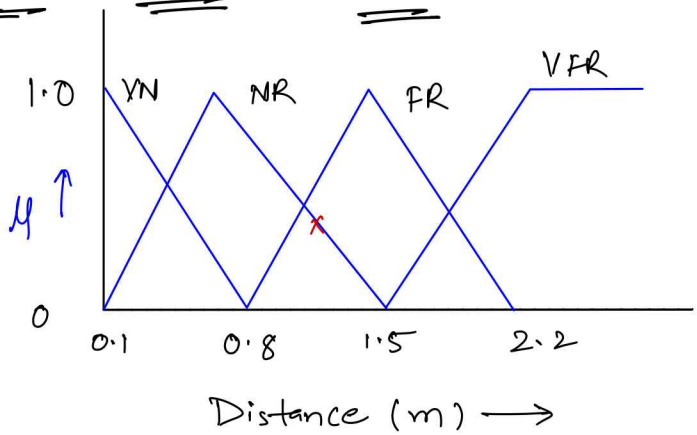
$$\mu_{Gaussian} = \frac{1}{e^{\frac{1}{2}\left(\frac{x-m}{\sigma}\right)^2}}$$

Navigation of Mobile Robot

Given Inputs: Distance = 1.04 m

Angle = 30°

For Distance = 1.04m



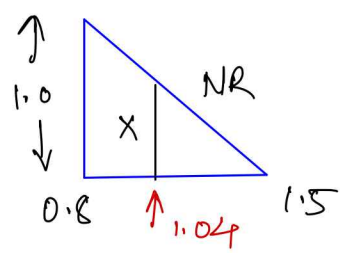
Step 1: Fuzzification.

⇒ Distance = 1.04 m is in between NR and FR

⇒ Therefore it may be either called NR (near) or FR (Far)

⇒ The membership values corresponding to μ_{NR} and μ_{FR} can be calculated from similar triangle rule.

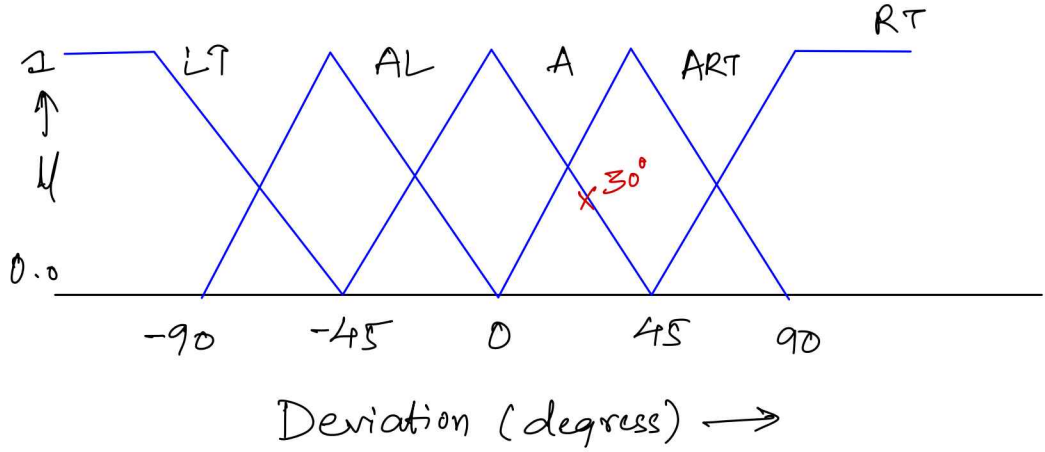
$$\boxed{\frac{x}{1.0} = \frac{1.5 - 1.04}{1.5 - 0.8}}$$



$\Rightarrow \quad \mu_{NR} = x = 0.6571$

$\Rightarrow \quad \mu_{FR} = 1 - \mu_{NR} = 0.3333.$

For Angle 30°



⇒ Similarly input angle of 30° may be A or ART

⇒ From similar triangle rule.

$\mu_A = 0.3333$

$\mu_{ART} = 0.6667$

Step 2: Inference Engine to find fired rules.

Out of 20 rules, only 4 rules are fired.

If Distance is NR AND Angle is A Then Deviation is RT

If Distance is NR AND Angle is ART Then Deviation is A

If Distance is FR AND Angle is A Then Deviation is ART

If Distance is FR AND Angle is ART Then Deviation is A

The strengths (α values) of fired rules are :

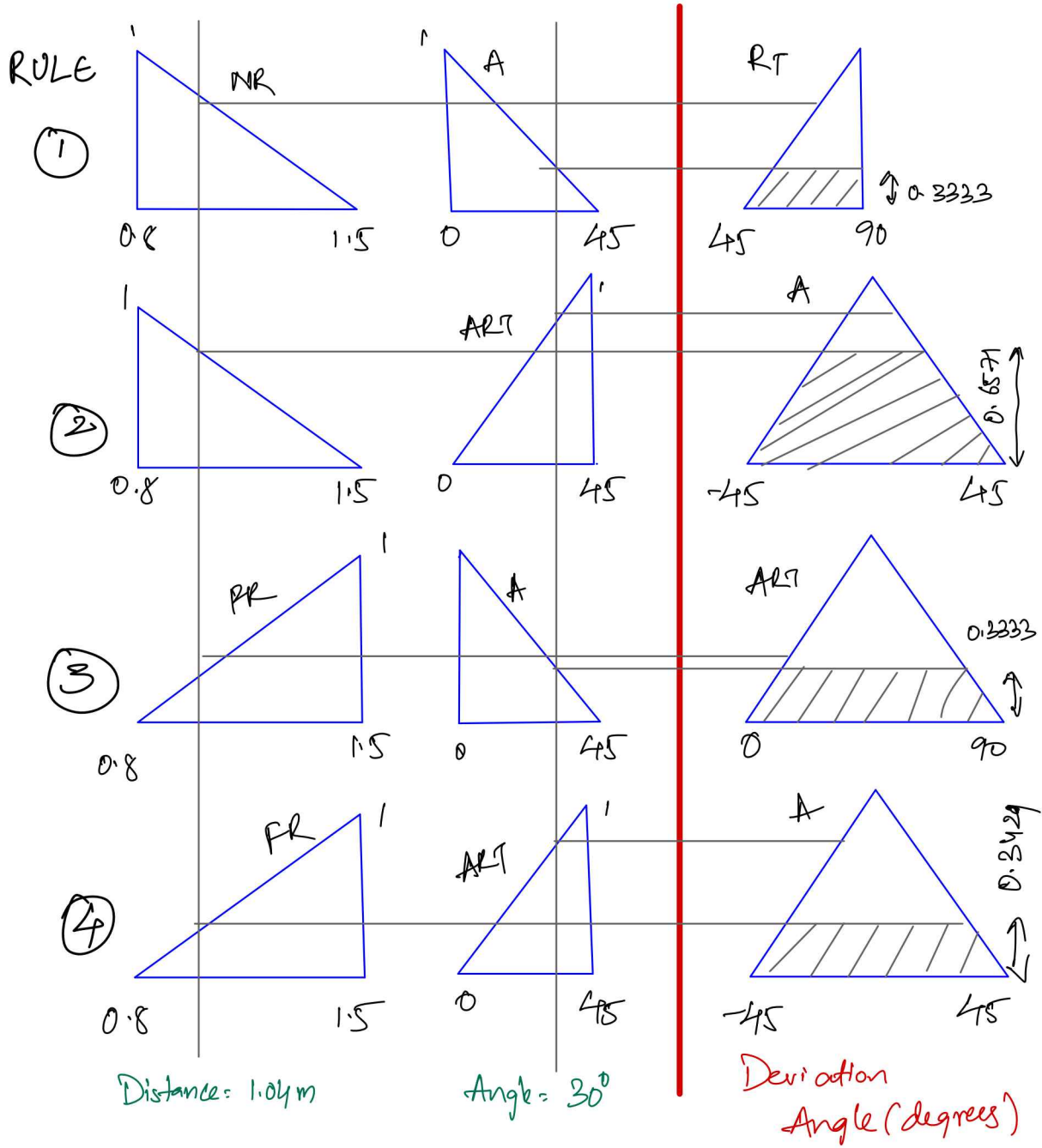
$\alpha_1 = \min(\mu_{NR}, \mu_A) = 0.3333$

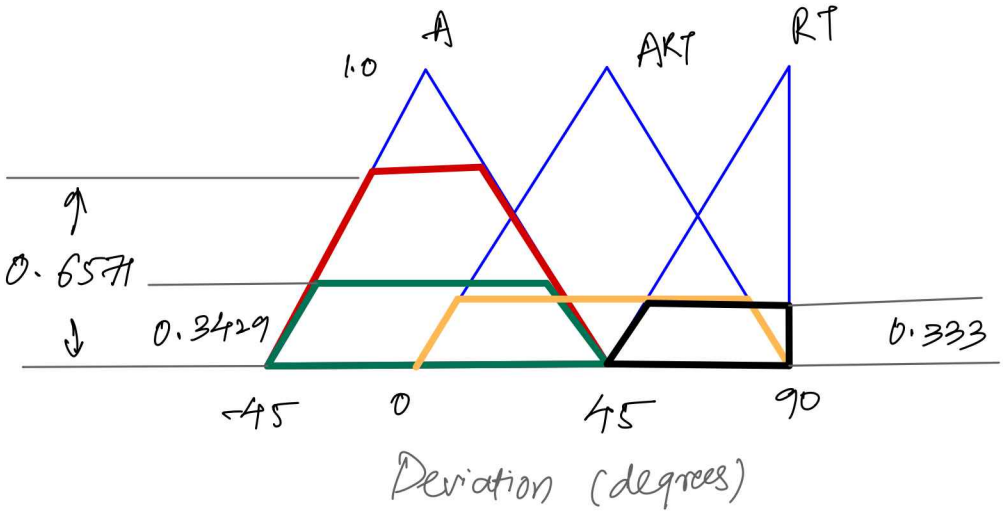
$\alpha_2 = \min(\mu_{NR}, \mu_{ART}) = 0.6571$

$\alpha_3 = \min(\mu_{FR}, \mu_A) = 0.3333$

$\alpha_4 = \min(\mu_{FR}, \mu_{ART}) = 0.3429$

The fuzzified Outputs corresponding to four different fired rules.





Combined fuzzified Output.

Step 3 : Defuzzification.

① Center of Sums method.

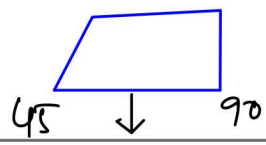
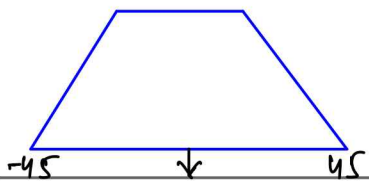
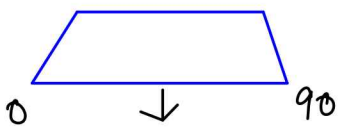
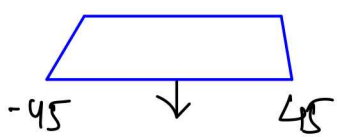
$$U'_{j'} = \frac{\sum_{j=1}^{p} A(\alpha_j) \times f_j}{\sum_{j=1}^{p} A(\alpha_j)}$$

$A(\alpha_j)$ → firing Area of jth rule

f_j → center of Area.

$U'_{j'}$ → Output of controller

Individualy calculate Area & center of area for each rule.

Fired Rule	Shaded Region	Area	Center
1		12.5	71
2		39.7019	0
3		25	45
4		25.5699	0

Area of Trapezoid

$$A = \frac{1}{2}(B_1 + B_2) \times H$$

B_1 → Base 1, B_2 → Base 2, H → Height

Centroid

$$C = \frac{B_1 + 2B_2}{3(B_1 + B_2)} \times H$$

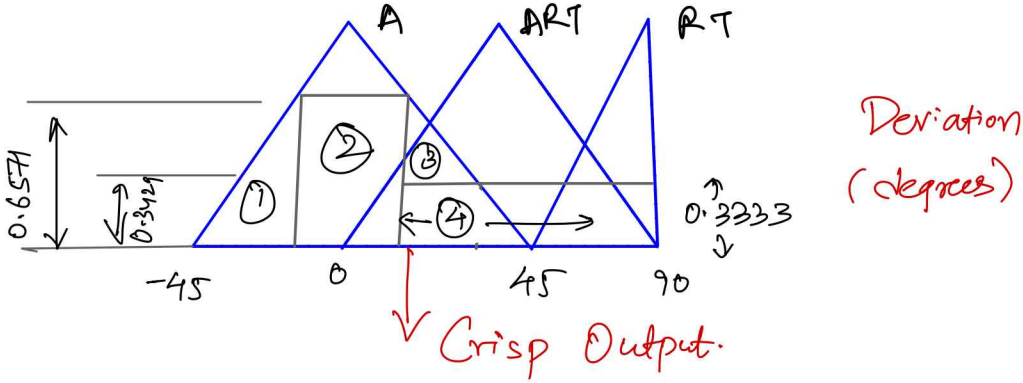
$$U = \frac{(12.5 \times 71) + (39.7089 \times 0) + (25 \times 45) + (25.6 \times 0)}{12.5 + 39.7089 + 25 + 25.5699}$$

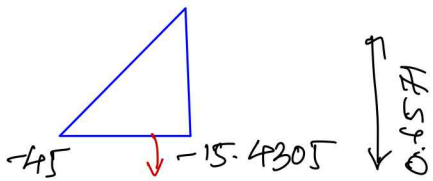
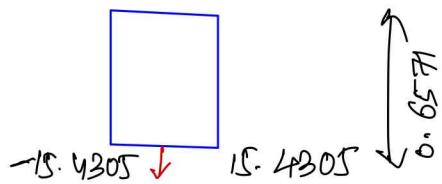
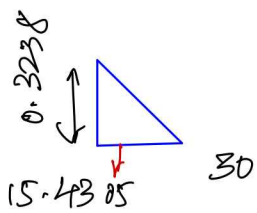
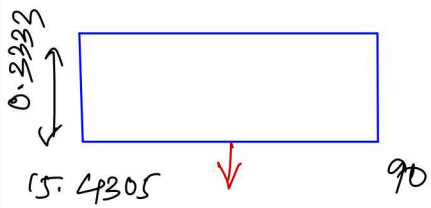
$$\boxed{U = 19.5809 \text{ degrees}} \rightarrow \text{Towards Right}$$

② Centroid Method

Centroid method is similar to center of sum method since it calculates Area and center of Area. (fast computing but less precise)

⇒ The key difference is instead of calculating Area & center for each output of a fired rule, we divide area and centre to known gemetry as simple triangle & rectangle and obtain output.



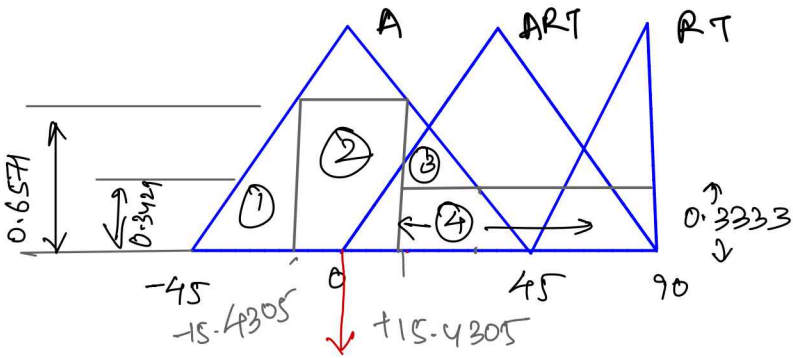
Marked Region	Shaded Region	Area	Center of Area
1		9.71	-25.28
2		20.27	0
3		2.35	20.28
4		24.85	52.71

$$U = \frac{(9.71 \times -25.28) + (20.27 \times 0) + (2.35 \times 20.28) + (24.85 \times 52.71)}{9.71 + 20.27 + 2.35 + 24.85}$$

$$\boxed{U = 19.4450}$$ → towards right.

③ Mean of Maxima Method.

⇒ Used when Output value is preferred as a crisp output (Integer)



Deviation (degrees)

⇒ Maximum value of membership is considered i.e. Area 2.

⇒ As the name suggests we need to calculate Mean of maximum membership value.

⇒ Range of Area = $\{-15.4305, 15.4305\}$

$$U = \text{Mean} = -15.4305 + 15.4305$$

$$\boxed{U = 0}$$

① Center of Sums → Precise but Computational cost is high

② Centroid → Fast calculation

③ Mean of Maxima → Integer Crisp Output is needed (Yes | No)